

Comparative Forecasting of Electric Vehicle Charging Demand Using Regression Models

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Abstract— This study examined the charging demands of electric vehicles (EV) over the past 10 years in a city and its effect on overall load patterns. The aim is to compare forecasting models and see which one offers the most reliable long term prediction of EV charging demand. With more people using EVs, it is important to estimate future charging behavior so that the grid can be managed properly. This creates a need to compare different forecasting methods in a structured way. In this paper a detailed comparative analysis of three regression-based forecasting models was conducted. Model performance was evaluated using standard metrics like the root mean squared error (RMSE). Our evaluation shows that bagged trees achieved the best performance (RMSE = 0.035), with the next best performance by fine trees (RMSE = 0.894) and robust linear (RMSE = 1.116). The forecasts point toward increasing charging demand up to 2035 stressing the need for effective grid planning, infrastructure planning and sustainable EV ecosystem development.

Keywords — *electric vehicle charging, regression model, comparative study, load forecasting, power station analysis*

I. INTRODUCTION

Electric vehicles are becoming more common and bring many practical benefits like reduced emissions and lower running costs. They are important for achieving sustainable transportation and offer savings through lower running costs, bring technological advancements in battery systems and energy storage. As EV numbers grow, it becomes difficult to maintain enough charging facilities that work productively. Thus, it is important to analyze and control the charging demand. In the last ten years, improvements in batteries and chargers have reduced EV charging time noticeably. These include many new technologies like DC fast charging, ultra-fast charging, higher power acceptance, standardization and compatibility, smart charging algorithms, over-the-air updates and improved efficiency. As battery technology develops and improves, the range of electric vehicles increases, high speed charging reduces downtime along the road. Drivers who travel long distances benefit more from EVs that can go farther on a single charge.

With the increasing adoption of electric vehicles (EVs) and the rising demand for commercial EV charging stations, many researchers have explored different ways to predict EV charging demand using a variety of modelling techniques. These forecasting methodologies can broadly be classified into four types: simulation models, time-series models, machine learning (ML) models, and deep learning (DL) models. In [2] Monte Carlo simulation to forecast the long-term charging load of plug-in EVs in China in 2015, 2020, and 2030 was applied. The authors accounted for various vehicle types such as electric buses, electric taxis, official cars and private cars etc., and used linear interpolation to estimate EV demand and charging patterns. They noted that charging patterns varied clearly between busy and non-busy hours. In recent years ML models are gaining popularity in EV charging demand forecasting due to their scalability and

robustness in handling large datasets. Such as in [3] where the next-day charging demand was predicted using a support vector machine (SVM), leveraging temporal features such as week-of-year, day-of-week and charging demand record of the same day-of-week of the previous four weeks. That study reported that the SVM-based model produced lower error values compared with the Monte-Carlo approach.

Other classical time series approaches have also been explored, such as in [4] seasonal autoregressive integrated moving average (SARIMA) was applied to predict aggregated power consumption from over 2,400 charging stations in Washington State and San Diego, California using two years of time-stamped data. Their tests show that SARIMA is better than ML models like random forest (RF) and gradient boosting regression tree (GBRT) in cases with limited training dataset. One drawback of older ML methods is that they do not handle time-dependent patterns very well. This can be solved by recurrent neural network (RNN) models, which are designed for sequential data processing and have been widely used in domains such as natural language processing, machine translation and time-series forecasting [5]. In [6] a short-term load forecasting for EV charging stations using deep learning approaches was conducted. Four different deep learning models, including deep neural network, standard RNN, Long Short-Term Memory LSTM and GRU were explored. Using 24 hours of charging load data, the models aimed to predict the load for the next hour. Results show that GRU delivered the highest predictive accuracy compared to the other models.

One study combined CNNs with LSTMs and found that mixing spatial feature extraction with temporal learning improved forecasting accuracy. They found that using both location-based features and time-related behavior improved the model's performance. Similarly, [8] another approach used Transformer models, which provided good accuracy and also trained faster than several earlier deep-learning methods. Recent analysis further convey the growing role of DL architectures such as Bi-LSTM, CNNs, and Transformers in capturing both spatial and temporal correlations in charging data [9].

In recent times, explainability has become an important factor. In [10], some researchers also applied SHAP-based explainability tools to tree-based models to understand which input factors influenced predictions the most. Similarly, [11] presented a systematic review of machine learning approaches for EV charging demand forecasting, highlighting research gaps and the need for interpretable, transferable models. In addition to these methods, newer studies have started exploring probabilistic and multi-level forecasting approaches. For instance, [12] applied deep transfer learning with multi-quantile temporal convolutional networks for probabilistic load forecasting, while [13] proposed forecasting at multiple levels of the charging

network, helping planners view demand trends region-wise. In cases where data availability is limited, transfer and meta-learning have been suggested. [14] showed how minimal data conditions can still bring robust short-term forecasts by leveraging pre-trained models and fine-tuning. Others have emphasized on other factors in demand forecasting, e.g., [15] integrated weather, traffic, routes, and charger location data to predict urban charging demand more accurately. Likewise, [16] combined CNN and XGBoost to model regional behaviors and spatial dependencies in smart cities. From a methodological perspective, [17] evaluated regression-based approaches including linear regression, decision trees, and ensemble methods within MATLAB's Regression Learner framework. Similarly, [18] compared time-series, ML, and DL techniques on EV load datasets, while [19] proposed multivariate ML frameworks (e.g., CatBoost, GRU) for charging station demand forecasting. These studies examined how each method balances accuracy, ease of explanation, and computational effort.

Thus, a comprehensive study that analyses and compares various forecasting models is needed for better data analysis and user acquisition. By the application of the Regression Learner tool, various forecasting models are obtained which are Ensemble Regression- Bagged Trees method, Linear Regression- Robust Linear method and Tree Regression-Fine Tree method. Based on these a comparative study for the forecast of charging demand is presented in Palo Alto city of California. Regression theory is applied for forecasting future energy consumption values based on its past values and increasing charging trends over time. The three models with least Root Mean Square Error have been applied to predict future value plots. Effective forecasts based on these models are necessary for analytical demands of electric vehicles. The plots share the analytical data for future preparation and for efficient decision making by policy-makers in the development of electric vehicles. In summary, forecasting charging data ensures a scalable, efficient, and user-friendly EV ecosystem. EVs are key in addressing environmental challenges and improving sustainable transportation systems. Predicting charging data is equally important for maximizing infrastructure, providing grid stability, reducing costs, and improving the overall user experience, making it a core of the evolution of electric mobility.

II. METHODOLOGY

As EV acceptance increases, trustworthy forecasting of charging demand becomes important. To address this need, this study uses a regression based method that highlights both forecasting accuracy and model transparency. This part explains the steps followed while developing and testing the models. It then details the training and validation of three models—Bagged Trees Ensemble, Robust Linear Regression, and Fine Trees Regression.

Regression Learner was used to train and compare the models as it helps plot errors and test different settings easily. The results for all the models are compared to see how different models work, and how charging demand changes over time. By comparing the models, we could see where it performs well and where it falls short.

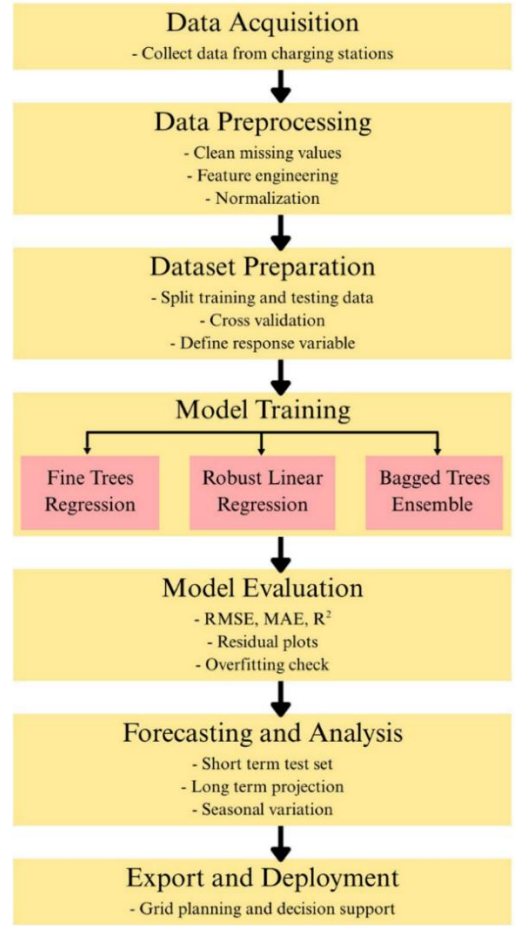


Fig. 1. Working flow of forecasting model

This study checks performance for the three models using RMSE, MAE, and R^2 along with residual. These metrics give different views of performance. RMSE shows the impact of large errors, MAE shows and measure of average error and R^2 shows how much of variation is explained by model. Checking the residuals also helped confirm if the models behaved uniformly for the whole dataset. An overview of the process is shown in Fig. 1. By using this method, the study aims to find the most effective forecasting method, and provide inputs that can support grid stability, infrastructure planning, and wider adoption of EVs. The next subsections describe each of the three forecasting models used in detail, outlining their theoretical foundations, implementation, and relevance to EV charging demand prediction.

A. Forecasting model 1: Bagged Trees Ensemble

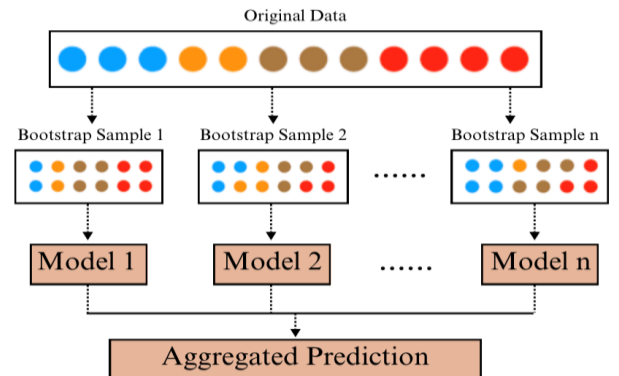


Fig. 2. Bagged Trees Ensemble

A regression tree ensemble is a predictive model composed of a weighted combination of multiple regression trees. The main idea behind the ensemble methodology is to aggregate multiple weighted models to obtain a combined model that outperforms every single model in it. Ensemble predictions generally require adjustment to remove biases and to ensure that the forecast frequency of events gives a realistic representation. Bagging and boosting are two main types of ensemble learning methods. Bagging, also known as Bootstrap Aggregating, is an ensemble method in machine learning that uses random subsets of the training data to train multiple models, and then combines their predictions. It is a part of parallel processing in the Ensembled Modelling Technique. It works by creating multiple random samples (bags) of the training data. Each decision tree is trained on these different bags and the feature space is partitioned into M disjoint regions $R_1^{(b)}, R_2^{(b)}, \dots, R_M^{(b)}$.

The prediction in region m for tree b given by $c_m^{(b)}$ is defined as shown in equation (1):

$$c_m^{(b)} = \frac{1}{|R_m^{(b)}|} \sum_{x_i \in R_m^{(b)}} y_i \quad (1)$$

where,

$R_m^{(b)}$ = m^{th} region of the b^{th} tree

$|R_m^{(b)}|$ = number of training samples in $R_m^{(b)}$

y_i = response value of training point x_i

Thus, the prediction for the b^{th} tree, denoted by $\hat{Y}^{(b)}(x)$ is given by equation (2):

$$\hat{Y}^{(b)}(x) = \sum_{m=1}^M c_m^{(b)} \cdot \mathbf{1}\{x \in R_m^{(b)}\} \quad (2)$$

(3) calculates the final bagged prediction $\hat{Y}(x)$ by averaging across all B trees:

$$\hat{Y}(x) = \frac{1}{B} \sum_{b=1}^B \hat{Y}^{(b)}(x) \quad (3)$$

Fig. 2 depicts the bootstrap aggregating methodology which helps to reduce variance, improve model stability, and leads to more accurate and robust predictions, especially when dealing with complex data.

B. Forecasting model 2: Robust Linear Regression

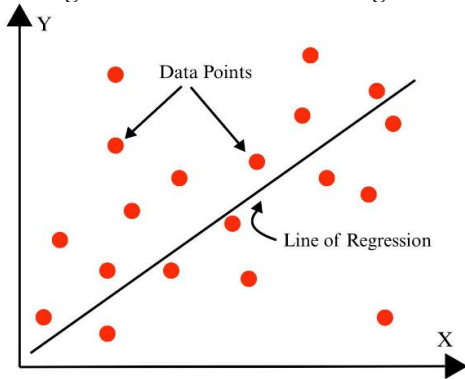


Fig. 3. Robust Linear Regression

Linear regression is a supervised learning algorithm that simulates a linear mathematical relationship between variables and makes predictions for continuous or numeric variables. Linear regression fits a straight line or surface that minimizes the discrepancies between predicted and actual output values as shown in Fig. 3. There are simple linear

regression calculators that use a “least squares” method to discover the best-fit line for a set of paired data. Robust linear regression is a method designed to solve the problems of traditional linear regression when data has outliers or does not follow basic assumptions such as normal distribution or equal variance. It reduces effect of outliers so they do not affect the results. It achieves this by using techniques like iteratively reweighted least squares, which assign different weights to data points, essentially down weighting outliers.

Modelling is done as shown in equation (4):

$$y_i = \beta_0 + \beta^T x_i + \varepsilon_i, \quad i = 1, 2, \dots, n \quad (4)$$

where,

y_i = response for sample i

β_0 = intercept

β = slopes for predictors (vector of regression coefficients)

ε_i = residual (error) for sample i

Unlike linear regression, in robust linear we minimize robust loss instead of the sum of squared errors. The robust loss is calculated as shown in equation (5):

$$\hat{\beta} = \arg \min_{\beta_0, \beta} \sum_{i=1}^n \rho\left(\frac{y_i - \beta_0 - \beta^T x_i}{\hat{\sigma}}\right) \quad (5)$$

where,

$\hat{\beta}$ = estimated regression coefficients

$\rho(\cdot)$ = robust loss function (like Huber, Tukey etc.)

$\hat{\sigma}$ = robust scale estimate of residuals

$\frac{y_i - \beta_0 - \beta^T x_i}{\hat{\sigma}}$ = standardized residual for sample i

This is solved by iteratively reweighted least squares (IRLS) as in (6).

$$\hat{\beta}^{(t+1)} = \arg \min_{\beta_0, \beta} \sum_{i=1}^n w_i^{(t)} (y_i - \beta_0 - \beta^T x_i)^2 \quad (6)$$

where,

$\hat{\beta}^{(t+1)}$ = coefficient estimates at iteration $t + 1$

$w_i^{(t)}$ = weight assigned to observation i at iteration t

$(y_i - \beta_0 - \beta^T x_i)^2$ = squared residual for observation i

the weights are calculated as shown in (7):

$$w_i^{(t)} = \frac{\varphi(u_i)}{u_i}, \quad u_i = \frac{y_i - \beta_0 - \beta^T x_i}{\hat{\sigma}}, \quad \varphi(u) = \rho'(u) \quad (7)$$

where,

u_i = standardized residual for observation i

$\varphi(u)$ = derivative of the robust loss function, also called the influence function

C. Forecasting model 3: Fine Trees Regression

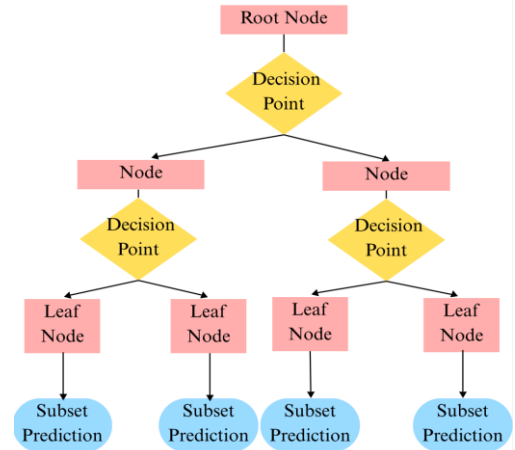


Fig. 4. Fine Trees Regression

Regression tree, is a ML technique used to predict continuous numerical outcomes based on input variables, that is the target variable can take continuous values rather than discrete class labels. Tree regression works by recursively partitioning data into subsets using input features that minimize prediction error at each node, creating a hierarchical structure from root to leaf nodes. In order to make a prediction for a given observation, we typically use the mean of the training data. This approach handles non-linear relationships without assuming data distributions, balancing interpretability with predictive power. The Fine Trees regression technique refers to a highly optimized regression tree model that uses highly detailed tree structure to make predictions and capture complex patterns in data. Fine Trees regression builds detailed predictive models by recursively splitting data into finer subsets using decision rules that minimize prediction error. It allows deeper tree structures with smaller leaf nodes to capture subtle patterns, controlled by hyperparameters. Each leaf predicts the mean target value of its subset; while pruning or regularization prevents overfitting by balancing granularity with generalization. Fig. 4 shows the working of the fine trees algorithm.

The regression tree partitions the feature space into M disjoint regions $R_1, R_2, R_3, \dots, R_M$

Modelling is done as according to equation (8):

$$\hat{f}(x) = \sum_{m=1}^M c_m \cdot \mathbf{1}\{x \in R_m\} \quad (8)$$

where,

$\hat{f}(x)$ = predicted value of regression tree for input x

c_m = predicted value assigned to region R_m

For squared error loss, the optimal constant c_m in each region R_m is the mean target value which is given in equation (9):

$$c_m = \frac{1}{|R_m|} \sum_{x_i \in R_m} y_i \quad (9)$$

where,

$|R_m|$ = number of training samples in region R_m

The tree grows greedily by choosing feature j and threshold t (as in equation (10)) such that it maximizes the reduction of sum of squared errors (SSE).

$$\Delta_{(j,t)} = SSE_{(s)} - [SSE_{(s_L)} + SSE_{(s_R)}] \quad (10)$$

where,

Δ = reduction in sum of squared errors (SSE)

S = current parent node

S_L, S_R = left and right child nodes after split

The SSE is calculated as:

$$SSE_{(A)} = \sum_{x_i \in A} (y_i - \bar{y}_A)^2 \quad (11)$$

where,

$\bar{y}_A$ = mean of all samples in region A

$(y_i - \bar{y}_A)^2$ = squared residual for sample i within region A

III. CASE STUDY

The input data used for the case study is from Palo Alto EV Charging Stations in California, United States. The data gives clear information on how charging stations are used in the city. The dataset has records of charging session durations, including the start and end times of sessions as shown by Table I.

Table I. EV Information of Real Data

EV ID	Start Date and Time	End Date and Time	Energy (kWh)	GHG Savings (kg)
3284	7/29/2011 20:17	7/29/2011 23:20	6.24945	2.625
4169	7/30/2011 0:00	7/30/2011 0:02	0.10658	0.045
.	.	.	.	.
804075	12/31/2020 17:49	12/31/202 0 21:55	11.976	5.03
2022651	12/31/2020 18:19	12/31/202 0 19:01	1.571	0.66

The energy consumption data gives information about the amount of electricity used during each session and also provides information on other parameters like station availability, usage rates, and demand over time. This information makes it possible to study usage trends, spot the busiest periods, and evaluate how well the charging stations operate.

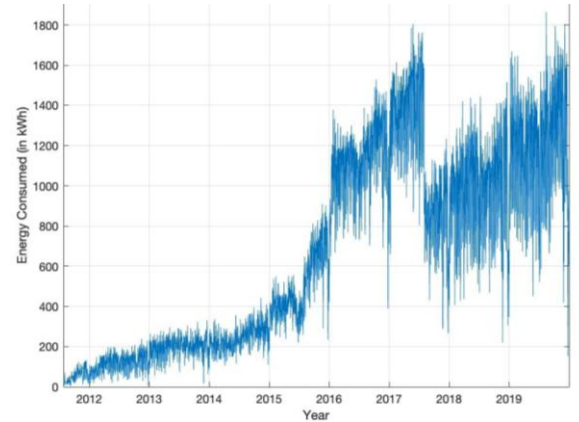


Fig. 5. Energy Consumption trends over the past decade

Using the initial dataset, seasonal variation in charging demand of electric vehicles at power stations of Palo Alto is generated as in Fig. 5. The plot suggests that charging demand remains fairly steady throughout the year with a few seasonal peaks. Mostly it is due to lower efficiency of batteries given the cold temperatures as the lithium-ion batteries consist of some chemical reactions which are slowed down and hence the range of electric vehicles is lowered, thus requiring more charge. Also, because batteries work best when warm, and some energy is used to heat them up before they can charge at full speed and hence increasing the charging time. The energy demand as observed in each month is shown in Fig. 6. with the help of a pie chart.

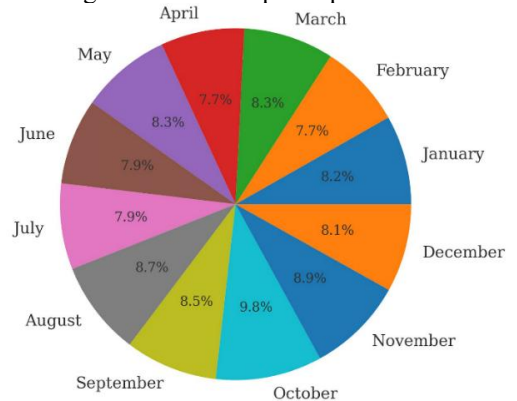


Fig. 6. Monthly distribution of EV charging energy consumption, showing higher loads in colder months.

IV. RESULTS

The models were trained and accordingly tested on the dataset for 2019. The actual vs predicted plots for various models has been shown in Fig. 7. The plots compare the predicted values with the actual data, letting us see how close each model came. A comparative analysis was also performed for realizing the superior model among the three as shown in Fig. 8. The most accurate result was given by Bagged Trees methodology of machine learning. The relative error and RMSE values were also compared for these models as in Table II.

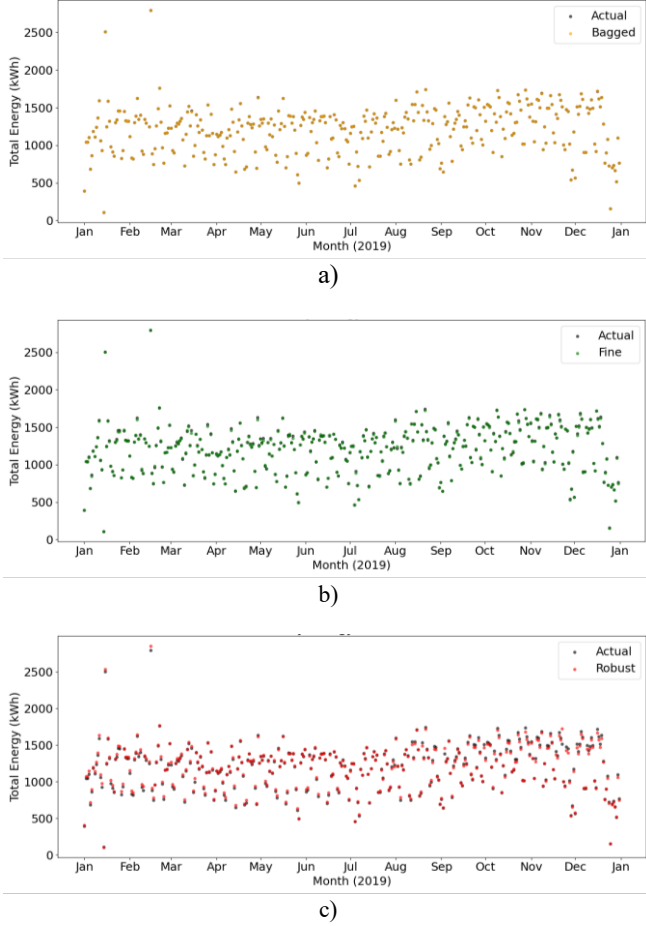


Fig. 7. Actual vs Predicted Energy Plot for Daily Energy Consumption using (a) Bagged Trees, (b) Robust Linear and (c) Fine Trees.

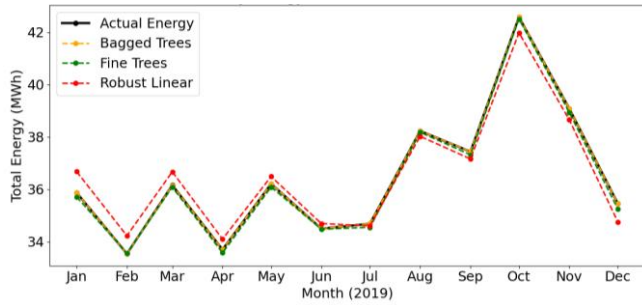


Fig. 8. Comparative analysis of actual vs predicted demand across the three models

Table II. Comparison of Evaluation Metrics

Model	RMSE (kWh)	MAE (kWh)	Relative Error (%)
Bagged Trees Ensemble	0.035	0.001	0.37
Fine Trees	0.894	0.563	9.55
Robust Linear	1.116	0.466	11.92

The trained models were able to produce reasonable forecasts for the coming decade as depicted in Fig. 9. It was observed that the average projections, indicated upward trends in energy requirements over the next decade. These forecasts help us in meeting the demands of the next decade. Planning ahead for future energy use is important for maintaining sustainable growth.

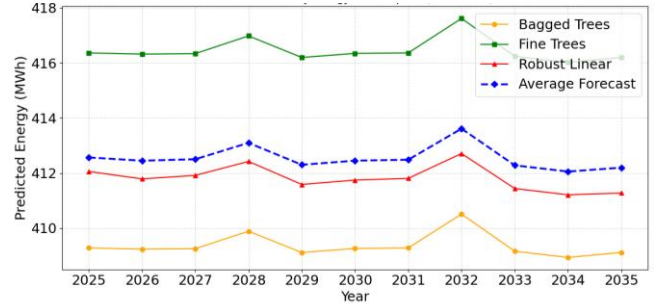


Fig. 9. Forecast of EV charging demand for 2025–2035

As stated earlier, the energy requirements also show a seasonal variation. Hence the forecast accounting for seasonal variations helps understand the energy requirements according to changing seasons. A forecast for the year 2030 with its seasonal variation has been presented in Fig. 10. It can be observed from the average forecast; the peaks are mostly observed in the colder months (Oct-Jan) similar to the data of the past decade.

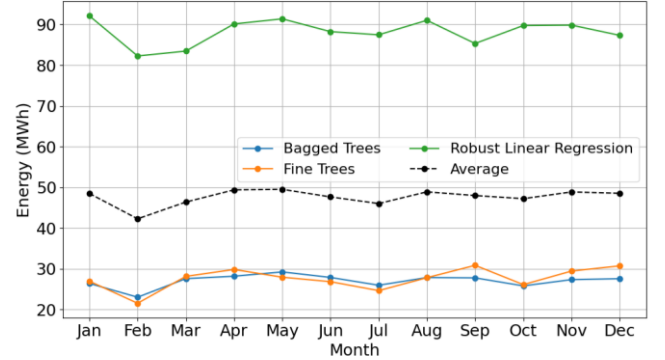


Fig. 10. Seasonal variation forecast for 2030

V. CONCLUSION

This paper presented a comparative evaluation of three regression-based forecasting techniques – bagged trees ensemble, fine trees regression, and robust linear regression – for estimating EV charging demand using real operational data from Palo Alto, California. The results show that bagged trees ensemble is consistently accurate across all evaluation metrics due to its ability to reduce variance and capture non-linear patterns in the data. Even though the fine trees model captured complex non linear demand variations, it was prone to overfitting whereas the robust linear model was less adaptive to dynamic charging patterns. Long term forecasting

from 2025-2035 indicates a significant upward trend in charging demand, with recurring seasonal peaks during winter months. The results show that predicting demand is important for planning power supply and infrastructure. Good forecast will help distribute energy in a better way. In the future, the study can be made better by adding more features like weather, traffic, and electricity prices. Future work can also look at hybrid and deep learning methods to reduce error and increase ability to scale.

REFERENCES

- [1] Z. Luo, Y. Song, Z. Hu, Z. Xu, X. Yang, and K. Zhan, "Forecasting charging load of plug-in electric vehicles in China," *Proc. IEEE Power Energy Soc. Gen. Meeting*, Detroit, MI, USA, 2011, pp. 1–8, doi: 10.1109/PES.2011.6039317.
- [2] E. S. Xydias, C. E. Marmaras, L. M. Cipcigan, A. S. Hassan, and N. Jenkins, "Forecasting electric vehicle charging demand using support vector machines," *Proc. Univ. Power Eng. Conf.*, Dublin, Ireland, 2013, pp. 1–6, doi: 10.1109/UPEC.2013.6714942.
- [3] H. M. Louie, "Time-series modelling of aggregated electric vehicle charging station load," *Electr. Power Compon. Syst.*, vol. 45, no. 14, pp. 1498–1511, 2017, doi: 10.1080/15325008.2017.1336583.
- [4] L. Medsker and L. C. Jain, *Recurrent Neural Networks: Design and Applications*. Boca Raton, FL, USA: CRC Press, 1999.
- [5] J. Zhu et al., "Short-term load forecasting based on deep learning," *Appl. Sci.*, vol. 9, no. 1, pp. 1–15, 2019.
- [6] Z. Shao et al., "A CNN-LSTM hybrid model for EV demand forecasting," *Appl. Energy*, vol. 311, pp. 118601, 2022, doi: 10.1016/j.apenergy.2022.118601.
- [7] Y. Wang et al., "Transformer-based approach for EV load prediction," *Sustainability*, vol. 15, no. 3, pp. 1125, 2023, doi: 10.3390/su15031125.
- [8] F. Ahmed, A. Karim, and S. Rahman, "Performance comparison of deep learning approaches in predicting EV charging demand," *Sustainability*, vol. 15, no. 5, pp. 4258, 2023, doi: 10.3390/su15054258.
- [9] L. Zhang et al., "Explainable EV demand forecasting using SHAP and XGBoost," *IEEE Access*, vol. 10, pp. 95374–95385, 2022, doi: 10.1109/ACCESS.2022.3205009.
- [10] M. Hamdaoui et al., "Machine learning-based electric vehicle charging demand forecasting: A systematized literature review," *Energies*, vol. 18, no. 17, pp. 4779, 2025, doi: 10.3390/en18174779.
- [11] J. Zhang, Y. Li, and H. Wu, "Location based probabilistic load forecasting of EV charging sites: Deep transfer learning with multi-quantile temporal convolutional network," *arXiv preprint*, arXiv:2409.11862, 2024.
- [12] M. O'Connell and K. Bhattacharya, "Coherent hierarchical probabilistic forecasting of electric vehicle charging demand," *arXiv preprint*, arXiv:2411.00337, 2024.
- [13] S. Zhao, A. El Gamal, and T. Zhang, "Short-term aggregate electric vehicle charging load forecasting in diverse conditions with minimal data using transfer and meta-learning," *Energy Syst.*, 2024, doi: 10.1007/s12667-024-00682-8.
- [14] H. Chen et al., "WeTRaC: A spatio-temporal forecasting model for EV charging demands integrating weather, traffic, routes and chargers," *Proc. IET Conf.*, 2025, doi: 10.1049/icp.2024.3747.
- [15] P. Sharma, V. Singh, and K. Nair, "Forecasting electric vehicle charging demand in smart cities using hybrid deep learning of regional spatial behaviours," *Energies*, vol. 18, no. 13, pp. 3425, 2025, doi: 10.3390/en18133425.
- [16] H. Khan et al., "Comparative analysis of regression models for EV load forecasting," *Sustainability*, vol. 15, no. 2, pp. 987, 2023, doi: 10.3390/su15020987.
- [17] T. Nguyen, S. Park, and D. Lee, "Short-term forecasting of electric vehicle load using time series, machine learning, and deep learning techniques," *World Electr. Vehicle J.*, vol. 14, no. 9, pp. 266, 2023, doi: 10.3390/wevj14090266.
- [18] Joshi and R. Patel, "Machine learning-based multivariate forecasting of electric vehicle charging station demand," *Electron. Lett.*, vol. 60, no. 1, pp. 21–24, 2024, doi: 10.1049/ell2.70104.